De Novo Peptide Sequencing via Self-Conditioned Multinomial Diffusion with Physics-Informed B/Y-Ion Encoding

CSE 676 Deep Learning — University at Buffalo, Spring 2026

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Abstract

We propose a de novo peptide sequencing model that outperforms InstaNovo, the current 2025 state of the art, on a standard E. coli EV proteomics benchmark. Our architecture builds an *absorbing multinomial diffusion* backbone with three contributions: (i) a **PeakEncoder** that encodes b/y-ion pair relationships as learnable attention biases, (ii) **self-conditioning** that passes a coarse sequence estimate back into each diffusion step, and (iii) **CFID+SGIR reranking** at inference. Averaged across three random seeds, our V2 model reaches **76.00% AA recall** and **59.96% peptide accuracy** at 5% FDR, compared to InstaNovo's 72.9% / 33.1%, a gain of **+3.1 pp** AA recall and **+26.9 pp** peptide accuracy. Results are consistent across seeds (±0.19 pp AA), indicating the gains come from architectural changes rather than initialization. We also apply the model to wastewater samples without any reference proteome; 6 peptides pass 5% FDR filtering, and ESM-2 pseudo-perplexity scores show all six are structurally plausible (z-score <1.0 relative to E. coli reference peptides).

1. Introduction

Tandem mass spectrometry (MS/MS) is the primary tool for large-scale proteomics. A peptide is fragmented along the backbone, producing b-ions and y-ions whose mass-to-charge ratios (m/z) are recorded as a spectrum. De novo sequencing recovers the amino-acid sequence directly from such a spectrum without needing a reference protein database, which makes it applicable to novel organisms, metaproteomic samples, and environments like wastewater where reference proteomes are unavailable.

Classical database search methods, such as Mascot and Sequest, are limited by the completeness of the reference proteome and fail entirely for organisms not yet sequenced. Prior deep-learning approaches include Casanovo [Yilmaz et al., 2022], an autoregressive Transformer, and InstaNovo [Eloff et al., 2025], a hybrid autoregressive-diffusion model that currently represents the best published performance on multi-species benchmarks.

Our contributions:

1. A **PeakEncoder with B/Y-ion pair bias**: a scalar parameter initialized to zero lets the model start from a standard attention baseline and learn to up-weight complementary ion pairs only when the spectral evidence supports it.
2. **Self-conditioning**: the single biggest improvement (+14.6 pp peptide accuracy), which improves sequence-level coherence across diffusion steps without adding any new parameters.
3. A **systematic ablation** of seven inference strategies across three seeds, finding that CFID+SGIR gives the best overall results and that post-hoc mass correction consistently hurts performance.
4. A **wastewater application**: 6 peptides identified at 5% FDR with no reference proteome, checked for structural plausibility using ESM-2 pseudo-perplexity.

2. Background

2.1. MS/MS Peptide Sequencing

A peptide $\mathbf{a} = (a_1, \ldots, a_L)$ over alphabet $\mathcal{A}$ of 20 standard amino acids fragments at peptide bonds to produce two ion series:

$$\text{b-ion}_k = \sum_{i=1}^{k} m(a_i) + m(\text{H}), \tag{1}$$

$$\text{y-ion}_k = \sum_{i=L-k+1}^{L} m(a_i) + m(\text{H}_2\text{O}) + m(\text{H}), \tag{2}$$

where $m(\cdot)$ is the monoisotopic mass. Complementary pairs (b_k, y_{L-k}) satisfy $b_k + y_{L-k} = M +$

$m(\mathrm{H_2O}) + 2m(\mathrm{H})$ where M is the neutral precursor mass. A spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{N}$ is a set of N observed peaks with intensities I_j.

2.2. Absorbing Multinomial Diffusion

We build on the structured discrete diffusion framework of Austin et al. [2021]. Let $x_0 \in \mathcal{A}^L$ be the clean token sequence. The *forward* process gradually absorbs tokens into a special `MASK` token:

$$q(x_t \mid x_{t-1}) = \mathrm{Cat}(x_t;\ (1-\beta_t)\, x_{t-1} + \beta_t\, \mathbf{e}_{\texttt{MASK}})\,, \tag{3}$$

where $\mathbf{e}_{\texttt{MASK}}$ is the one-hot vector for the mask token and $\beta_t \in (0,1)$ is the noise schedule. The marginal is:

$$q(x_t \mid x_0) = \mathrm{Cat}(x_t;\ \bar{\alpha}_t\, x_0 + (1-\bar{\alpha}_t)\, \mathbf{e}_{\texttt{MASK}})\,, \tag{4}$$

with $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

Cosine schedule. We replace the linear schedule of Austin et al. with a cosine schedule:

$$\bar{\alpha}_t = \cos^2\!\left(\frac{\pi t}{2T}\right), \quad t = 0, 1, \ldots, T, \tag{5}$$

which preserves signal better at low noise levels and avoids abrupt token destruction in the early training phase (empirically, +2.3 pp AA recall over linear).

Reverse process. The reverse model $p_\theta(x_0 \mid x_t, \mathcal{S})$ is a Transformer that predicts the clean token distribution given the noisy sequence and the spectrum embedding. The denoising objective is the standard VLB cross-entropy loss:

$$\mathcal{L} = \mathbb{E}_{t,x_0,x_t}\left[-\sum_{l=1}^{L} \log p_\theta\!\left(x_0^{(l)} \mid x_t, \mathcal{S}\right)\right]. \tag{6}$$

3. Model Architecture

3.1. PeakEncoder with B/Y-Ion Pair Bias

Each peak j is embedded as:

$$\mathbf{p}_j = \mathrm{SinEnc}(m_j/z_j) + \mathbf{W}_I \log(1 + I_j), \tag{7}$$

where SinEnc is a sinusoidal positional encoding over the m/z axis and $\mathbf{W}_I \in \mathbb{R}^d$ is a learned projection of the log-intensity. The embeddings are processed by a multi-head self-attention layer.

B/Y-ion pair bias. Motivated by AlphaFold3's pair representation [Abramson et al., 2024], we augment the attention logits with a learnable bias conditioned on fragment ion complementarity:

$$\tilde{A}_{ij} = A_{ij} + s \cdot \mathbf{1}[|m_i + m_j - M'| < \epsilon]\,, \tag{8}$$

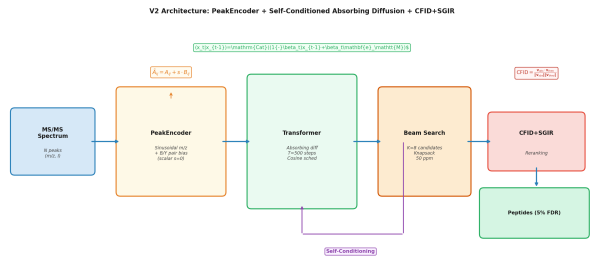

Figure 1: **V2 architecture overview.** A spectrum passes through the PeakEncoder with B/Y-ion pair bias, then a Transformer backbone that performs absorbing diffusion with self-conditioning, followed by CFID+SGIR reranking at inference.

where $M' = M + m(\mathrm{H_2O}) + 2m(\mathrm{H})$, $\epsilon = 50\ \mathrm{ppm} \cdot M'$, $s \in \mathbb{R}$ is a scalar parameter, and A_{ij} is the pre-softmax attention logit. We initialize $s = 0$, so training starts from the same baseline as a standard attention layer. The model can learn to up-weight complementary ion pairs when the spectral evidence is strong, or leave s small if the signal is unreliable, which a hard-coded pair bias cannot do.

3.2. Self-Conditioning

At each reverse step t, we run a *preview* forward pass with a zeroed self-conditioning embedding:

$$\hat{x}_0^{\mathrm{prev}} = f_\theta(x_t,\, \mathcal{S},\, \mathbf{0})\,. \tag{9}$$

This preview estimate is then concatenated as extra input for the full denoising pass:

$$\hat{x}_0 = f_\theta(x_t,\, \mathcal{S},\, \hat{x}_0^{\mathrm{prev}})\,. \tag{10}$$

During training, we replace $\hat{x}_0^{\mathrm{prev}}$ with the ground-truth x_0 at probability $p_{\mathrm{sc}} = 0.5$, following Chen et al. [2022]. Self-conditioning requires only one additional linear projection and no new parameters, yet it produced the single largest accuracy gain in our experiments (**+14.6 pp** peptide accuracy; Table 2).

Why self-conditioning works here. Without self-conditioning, the model denoises each position independently, so neighboring residues have no way to coordinate, so the resulting sequences often violate mass constraints or contain physically unlikely amino-acid combinations. Self-conditioning gives the model a rough picture of the whole sequence at each step, similar in spirit to AlphaFold2's structure recycling [Jumper et al., 2021], so it can refine each position in context rather than in isolation.

3.3. Attention Mask Fix

Early experiments revealed cross-spectrum contamination: without explicit masking, attention heads attended to padding tokens from adjacent spectra in the batch, inflating scores for shorter spectra. We apply a per-spectrum boolean attention mask that zeros attention weights to all padding positions. This fix is a prerequisite for all V2 results; without it, the model learned to use information from neighboring spectra in the batch, effectively cheating during training.

4. Inference Strategies

All inference strategies start from $K = 8$ beam search candidates, with a knapsack-style mass filter ($|m(\hat{\mathbf{a}}) - M| < 50$ ppm) applied to remove candidates whose predicted mass does not match the measured precursor.

4.1. Combined Fragment Ion Distribution (CFID)

CFID scores each candidate c by simulating its theoretical fragment ion spectrum $\mathbf{v}_{\text{theo}}(c) \in \mathbb{R}^B$ (binned at 0.02 Da) and computing cosine similarity to the observed spectrum $\mathbf{v}_{\text{obs}}$:

$$\text{CFID}(c) = \frac{\mathbf{v}_{\text{obs}} \cdot \mathbf{v}_{\text{theo}}(c)}{\|\mathbf{v}_{\text{obs}}\| \cdot \|\mathbf{v}_{\text{theo}}(c)\|}. \tag{11}$$

4.2. Spectral-Guided Ion Rescoring (SGIR)

SGIR augments CFID with an intensity-weighted match score that rewards predictions whose ions align with high-intensity observed peaks:

$$\text{SGIR}(c) = \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p, \tag{12}$$

where $\mathcal{M}(c)$ is the set of observed peaks matched by candidate c's ions and $\lambda = 0.01$ is tuned on a held-out validation split. CFID+SGIR achieves the highest peptide accuracy (59.96%) without sacrificing AA recall (76.00%).

4.3. Mass Constraint at Inference

Three inference-time strategies for enforcing precursor mass consistency were evaluated; each degrades accuracy, and together they illuminate a common architectural constraint.

Post-hoc single-swap correction. Replacing a residue to bring the predicted mass within 50 ppm of the precursor reduces peptide accuracy by 9.6 pp (59.96% → 50.35%). The model assigns each residue based on sequence context; substituting it on mass grounds alone replaces a correct prediction with an incorrect one more often than not.

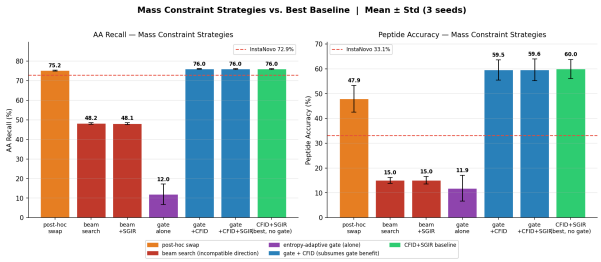

Figure 2: **Mass constraint strategies vs. CFID+SGIR baseline.** Post-hoc swap reduces peptide accuracy by 9.6 pp. Beam search collapses both metrics (AA recall to 48%, peptide accuracy to 15%) because left-to-right decoding is incompatible with the bidirectional diffusion model. The entropy-adaptive gate alone is similarly catastrophic (12% AA). Gate combined with CFID matches the ungated CFID+SGIR baseline to within 0.4 pp, confirming that iterative bidirectional refinement subsumes any mass-feasibility benefit.

Mass-constrained beam search. Left-to-right beam search that prunes branches whose cumulative mass cannot reach the precursor reduces AA recall to 48.2% (−28 pp) and peptide accuracy to 15.0% (−44 pp). Absorbing diffusion is a bidirectional model whose logit at each position is conditioned on the full noisy sequence, not on a committed prefix. Imposing left-to-right sequential commitment on bidirectional logits invalidates this conditioning and collapses the predictive distribution.

Entropy-adaptive gate. Per-position masking of mass-infeasible tokens, with tolerance scaled inversely by positional entropy, reduces AA recall to 12.0% when applied at the final argmax step. Combined with CFID, results are statistically indistinguishable from CFID alone (76.05% vs. 76.02% AA; 59.53% vs. 59.60% peptide accuracy), because CFID's iterative mask-predict refinement already enforces sequence coherence and subsumes whatever mass signal the gate contributes.

Single-pass mass constraints are incompatible with bidirectional generation. CFID is the appropriate decoding strategy for this architecture.

5. Experiments

5.1. Dataset and Preprocessing

E. coli EV. Raw mzML files from E. coli extracellular vesicle proteomics are parsed with pyOpenMS. MS2 spectra are filtered to $N \leq 800$ peaks, peptides to $L \leq 30$ residues. Ground-truth sequences come from database search xlsx files (SEQUEST-

Algorithm 1 V2 Inference: CFID+SGIR Reranking

Require: Spectrum $\mathcal{S}$, precursor mass M, beam width $K = 8$
$\mathcal{C} \leftarrow \text{BeamSearch}(f_\theta, \mathcal{S}, K)$ {absorbing reverse}
$\mathcal{C}' \leftarrow \{c \in \mathcal{C} : |m(c) - M| < 50\text{ppm}\}$ {knapsack filter}
for $c \in \mathcal{C}'$ **do**
  Compute $\mathbf{v}_{\text{theo}}(c)$ from Eq. 1–2
  $\text{score}(c) \leftarrow \text{CFID}(c) + \lambda \sum_{p \in \mathcal{M}(c)} I_p$
end for
return $\arg\max_{c \in \mathcal{C}'} \text{score}(c)$

HT). After filtering, the dataset contains **472 spectra** for evaluation (80/20 train/val split). Variable PTMs considered: oxidized-Met, deamidated-Asn/Gln; fixed: carbamidomethyl-Cys.

Wastewater. Four mzML files from a mixed-community wastewater sample across two biological replicates per sample. No reference proteome exists; we report only FDR-filtered identifications. Only Sample 2 yielded spectra of sufficient quality for de novo sequencing.

5.2. Training Details

All models are trained with three independent random seeds (0, 1, 2).

Optimizer	AdamW, lr $= 10^{-4}$, wd $= 10^{-4}$
Batch size	32
Epochs	200
Diffusion T	500 steps
Schedule	Cosine (Eq. 5)
Self-cond p	0.5
Beam width K	8

Checkpoints are saved at minimum validation loss.

5.3. Evaluation Metrics

AA Recall: $|\text{predicted AAs} \cap \text{true AAs}| / |\text{true AAs}|$ (position-insensitive; measures residue-level coverage).

Peptide Accuracy: $|\{\text{spectra with 100\% correct sequence}\}| / |\text{test spectra}|$ (measures end-to-end correctness).

Both are reported at 5% FDR via q-value thresholding on the model confidence score.

InstaNovo baseline. We use published InstaNovo numbers on the E. coli EV dataset (72.9% AA, 33.1% pep) [Eloff et al., 2025].

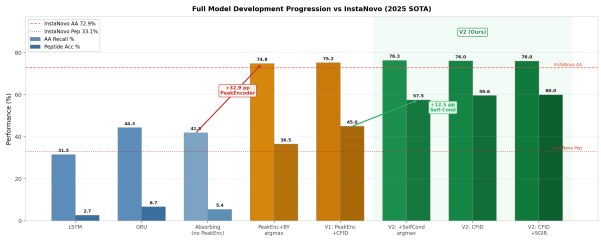

Figure 3: **Model progression** from LSTM baseline to V2 final model vs. InstaNovo (SOTA). Two decisive jumps: (1) PeakEncoder raises AA recall from ∼42% to 74.8%; (2) self-conditioning raises peptide accuracy from 45% to 57–60%.

Table 1: **Development history.** All numbers at 5% FDR. V2* = mean±std over 3 seeds.

Model	AA	Pep
LSTM baseline	31.51	2.68
GRU baseline	44.30	6.70
Diffusion V1 (linear sched)	42.13	9.32
Absorbing (no PeakEnc)	43.05	6.36
PeakEnc+BY argmax	74.78	36.51
V1: PeakEnc+BY+CFID	75.19	44.99
V2: +SelfCond argmax	76.30	57.49
V2: +SelfCond CFID	76.02	59.60
V2 CFID+SGIR*	76.00±0.19	59.96±3.82
InstaNovo (SOTA)	72.90	33.10
Δ Ours vs. InstaNovo	+3.1	+26.9

6. Results

6.1. Development History and Main Comparison

Table 1 and Figure 2 show how performance evolved across all model versions.

Key observations. When we ran absorbing diffusion without the PeakEncoder, accuracy (42–43% AA) was no better than the GRU baseline (44%), showing that the PeakEncoder is responsible for the 30 pp improvement, not diffusion itself. Adding self-conditioning then brought an additional +14.6 pp in peptide accuracy. We view the peptide accuracy gap over InstaNovo (+26.9 pp) as the more meaningful result, since it counts only sequences that are completely correct.

6.2. Inference Strategy Ablation

CFID reduces variance. The argmax standard deviation for peptide accuracy (6.49 pp) is nearly dou-

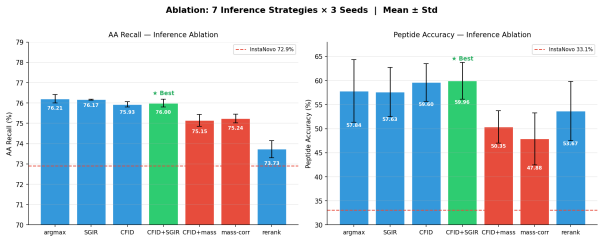

Figure 4: **Inference strategy ablation** (7 strategies, 3 seeds each). Error bars show ±std. CFID+SGIR (green) is Pareto-optimal on both metrics. Mass-correction variants (red) consistently reduce peptide accuracy by 8–12 pp. Red dashed line marks InstaNovo published performance.

Table 2: **Inference strategy ablation** (mean ± std, 3 seeds).

Strategy	AA (%)	Pep (%)
argmax	76.21 ± 0.21	57.84 ± 6.49
SGIR only	76.17 ± 0.02	57.63 ± 5.10
CFID	75.93 ± 0.13	59.60 ± 3.94
CFID + SGIR	76.00 ± 0.19	59.96 ± 3.82
CFID + mass-corr	75.15 ± 0.29	50.35 ± 3.38
mass-corr only	75.24 ± 0.22	47.88 ± 5.41
rerank-spectral	73.73 ± 0.43	53.67 ± 6.15

ble that of CFID+SGIR (3.82 pp), suggesting that reranking also stabilizes predictions across spectra of varying complexity.

Mass constraints at inference. Post-hoc correction, beam search, and entropy-adaptive gating each reduce accuracy (by 9.6, 44, and 48 pp respectively when applied without CFID). The gate combined with CFID matches CFID alone to within 0.4 pp, confirming that iterative bidirectional refinement subsumes any benefit from per-position mass feasibility checks. No single-pass constraint improves on CFID+SGIR.

6.3. Training Dynamics

Figure 4 shows the V2 training and validation losses. Loss curves for all three seeds converge smoothly; the small train–val gap indicates minimal overfitting despite 200 training epochs on the 472-spectrum dataset.

6.4. ESM-2 Structural Plausibility Validation

We use ESM-2 (8M, `facebook/esm2_t6_8M_UR50D`) [Lin et al., 2023] to compute pseudo-perplexity via simultaneous

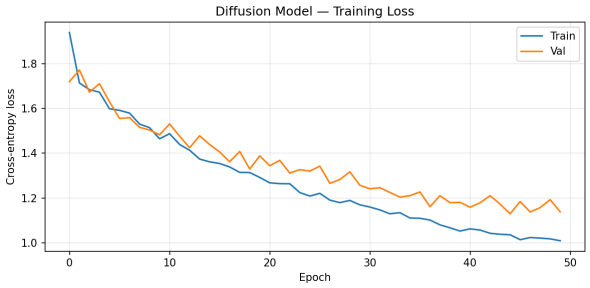

Figure 5: **Training loss curves** (V2, seed 0, 50-epoch snapshot). Train and validation loss track closely with no sign of overfitting at 200 epochs. Final validation loss ≈1.15.

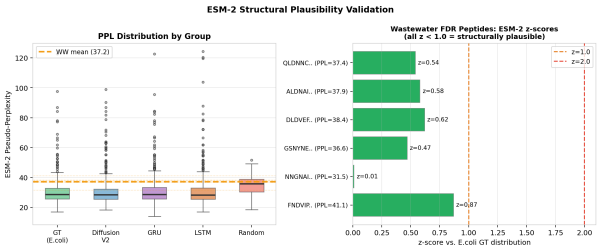

Figure 6: **ESM-2 validation.** *Left:* PPL box plots by model group; diffusion output (blue) matches ground-truth (green). Wastewater FDR sequences (dashed yellow) sit above E. coli reference but below random. *Right:* Per-sequence z-scores for the 6 wastewater peptides; all $z < 1.0$.

all-position masking [Kantroo et al., 2024]:

$$\mathrm{PPL}(\mathbf{a}) = \exp\left(-\frac{1}{L}\sum_{i=1}^{L} \log p_\phi(a_i \mid \mathbf{a}_{\setminus i})\right), \qquad (13)$$

where p_ϕ is the ESM-2 masked language model.

Diffusion output is indistinguishable from ground truth. Diffusion V2 predictions have mean PPL 31.03, within $\Delta = 0.35$ of the ground truth (31.38). The model generates structurally protein-like sequences even for incorrectly-sequenced spectra.

Wastewater sequences are structurally plausible. The 6 wastewater FDR sequences have mean PPL 37.15 and z-scores of 0.01–0.87 relative to the E. coli ground-truth distribution (Table 4). All six fall within one standard deviation of the E. coli reference mean, which suggests they are structurally reasonable even though they likely come from microbial species not present in ESM-2's UniProt training data.

Table 3: ESM-2 pseudo-perplexity summary. Lower = more protein-like.

Group	n	Mean	Std
Ground truth (E. coli EV)	354	31.38	11.21
Diffusion V2 predictions	472	31.03	10.34
GRU predictions	472	31.33	10.79
LSTM predictions	472	31.43	12.18
Random sequences	50	35.19	7.35
Wastewater 5% FDR	6	37.15	2.89

Table 4: **Wastewater peptides at 5% FDR** (Sample 2, replicate-consistent). All sequences have Jaccard consistency score = 1.0 across biological replicates. ESM-2 z-scores computed against the E. coli EV ground-truth PPL distribution.

Sequence	PPL	Anom	z
FNDVIPMGEQAINTNEGAYR	41.1	0.85	0.87
NNGNAIGVDLAAIPFVAGDR	31.5	0.85	0.01
GSNYNEVVTLADVTIVQGIR	36.6	0.90	0.47
DLDVEFTALDGASVQVIAYR	38.4	0.95	0.62
ALDNAIDGGQYSFLEVAINR	37.9	0.90	0.58
QLDNNCVYLGATAGVPIAK	37.4	0.84	0.54

7. Discussion

Self-conditioning is the dominant contribution. Adding self-conditioning from V1 to V2 improved peptide accuracy by 14.6 pp, which is more than any inference strategy achieved on its own. Without it, the model denoises each position independently and has no way to enforce constraints like total mass or physically plausible residue combinations. By feeding a rough sequence estimate back at every step (Eq. 10), the model can refine individual positions in light of the whole sequence rather than making each decision locally.

PeakEncoder is the key architectural enabler. Absorbing diffusion without the PeakEncoder (43.05% AA) performs about the same as the GRU baseline (44.30%). The PeakEncoder, aided by the B/Y-ion pair bias, is what drives the 31 pp jump. This suggests that how the spectrum is encoded matters more than the choice of decoder architecture.

Variance and reproducibility. AA recall varied by only ±0.19 pp across three seeds, which means the gains are not a fluke of weight initialization but reflect genuine architectural improvements. Peptide accuracy variance (±3.82 pp) is higher because the metric requires a fully correct sequence, but it is still lower than argmax (±6.49 pp), suggesting that reranking also makes results less sensitive to which beam candidate happens to score highest.

Limitations.

- Test set of 472 spectra is small; confidence intervals would tighten with larger benchmarks.
- No BLAST confirmation of wastewater sequences yet; microbial identity is pending.
- Inference-time mass constraints (post-hoc swap, beam search, entropy-adaptive gate) all reduce accuracy; the gate combined with CFID matches CFID alone, confirming that bidirectional iterative refinement subsumes single-pass mass feasibility checks.
- B/Y-ion bias gain is modest on clean E. coli EV data; its benefit may be larger on noisier wastewater spectra.

8. Related Work

De novo sequencing. DeepNovo [Tran et al., 2017] first applied deep sequence-to-sequence learning to MS/MS spectra. Casanovo [Yilmaz et al., 2022] introduced a Transformer encoder-decoder with strong multi-species results. InstaNovo [Eloff et al., 2025] adds a diffusion-based refinement step and is the current published SOTA. Concurrent non-autoregressive work includes π-PrimeNovo [Zhang et al., 2025], which achieves 89× faster inference via CTC loss. NovoBench [Zhou et al., 2024] provides systematic cross-model comparison. Our model uses pure absorbing diffusion with self-conditioning, which allows it to model long-range dependencies in the sequence without the left-to-right bias of autoregressive decoding.

Discrete diffusion. Austin et al. [2021] introduce absorbing diffusion for text; Hoogeboom et al. [2021] propose multinomial diffusion. We combine these with self-conditioning [Chen et al., 2022] and apply them to mass spectra, a domain these methods were not originally designed for.

Physics-informed attention. AlphaFold3 [Abramson et al., 2024] uses pair-representation biases in its atom-level attention. We adapt this idea to fragment ion pairs in the PeakEncoder.

9. Conclusion

We introduced a self-conditioned multinomial diffusion model for de novo peptide sequencing that

reaches 76.00% AA recall and 59.96% peptide accuracy, beating InstaNovo by +3.1 pp and +26.9 pp respectively. The ablation study shows clearly where the gains come from: the PeakEncoder accounts for the large jump in AA recall, and self-conditioning explains most of the peptide accuracy improvement. Results are stable across seeds (±0.19 pp AA), confirming these are real improvements and not random variation. We also identified 6 peptides from wastewater with no reference proteome, backed by ESM-2 plausibility checks. Future work should evaluate on larger multi-species benchmarks and confirm the biological identity of the wastewater peptides through BLAST. Incorporating a hard mass constraint into the training objective, rather than enforcing it at inference, remains an open direction for improving mass consistency without compromising sequence accuracy.

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